

HD74HC354

8-to-1-line Data Selector/Multiplexer/Register (with 3-state outputs)

REJ03D0613-0200
(Previous ADE-205-492)
Rev.2.00
Jan 31, 2006

Description

This data selectors/multiplexers contains full on-chip binary decoding to select one of eight data sources. The data select address is stored in transparent latches that are enabled by a low level address on pin 11, Select Control. Data on the 8 input lines is stored in a parallel input/output register which in the HD74HC354 is composed of 8 transparent latches enabled by a low level on pin 9, Data Control. Both true (Y) and complementary (W) 3-state outputs are available.

Features

- High Speed Operation: t_{pd} ($\overline{DC}$ to W, Y) = 23 ns typ ($C_L = 50$ pF)
- High Output Current: Fanout of 15 LSTTL Loads
- Wide Operating Voltage: $V_{CC} = 2$ to 6 V
- Low Input Current: 1 μ A max
- Low Quiescent Supply Current: I_{CC} (static) = 4 μ A max ($T_a = 25^\circ\text{C}$)
- Ordering Information

Part Name	Package Type	Package Code (Previous Code)	Package Abbreviation	Taping Abbreviation (Quantity)
HD74HC354P	DILP-20 pin	PRDP0020AC-B (DP-20NEV)	P	—
HD74HC354FPEL	SOP-20 pin (JEITA)	PRSP0020DD-B (FP-20DAV)	FP	EL (2,000 pcs/reel)
HD74HC354RPEL	SOP-20 pin (JEDEC)	PRSP0020DC-A (FP-20DBV)	RP	EL (1,000 pcs/reel)

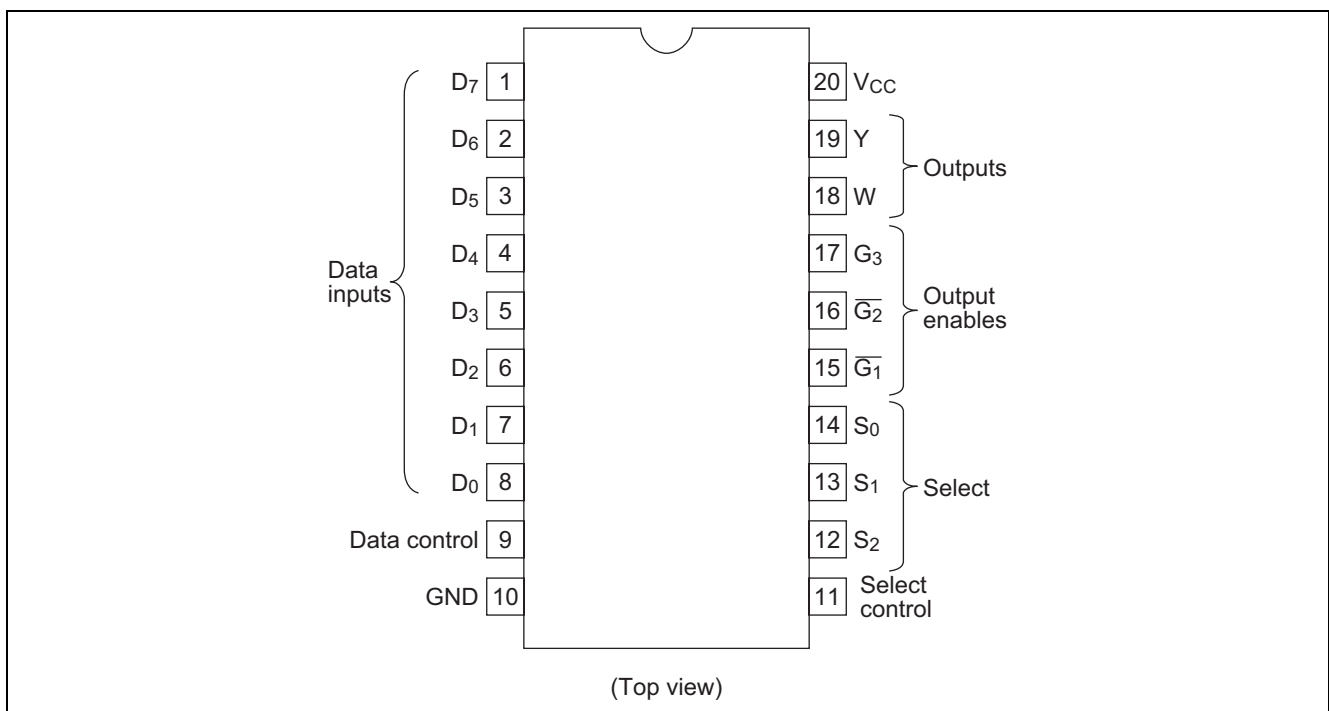
Note: Please consult the sales office for the above package availability.

Function Table

Inputs							Outputs	
Select			Data Control	Output Enable				
S ₁	S ₂	S ₀	DC	G ₁	G ₂	G ₃	W	Y
X	X	X	X	H	X	X	Z	Z
X	X	X	X	X	H	X	Z	Z
X	X	X	X	X	X	L	Z	Z
L	L	L	L	L	L	H	D ₀	D ₀
L	L	L	H	L	L	H	D _{0n}	D _{0n}
L	L	H	L	L	L	H	D ₁	D ₁
L	L	H	H	L	L	H	D _{1n}	D _{1n}
L	H	L	L	L	L	H	D ₂	D ₂
L	H	L	H	L	L	H	D _{2n}	D _{2n}
L	H	H	L	L	L	H	D ₃	D ₃
L	H	H	H	L	L	H	D _{3n}	D _{3n}
H	L	L	L	L	L	H	D ₄	D ₄
H	L	L	H	L	L	H	D _{4n}	D _{4n}
H	L	H	L	L	L	H	D ₅	D ₅
H	L	H	H	L	L	H	D _{5n}	D _{5n}
H	H	L	L	L	L	H	D ₆	D ₆
H	H	L	H	L	L	H	D _{6n}	D _{6n}
H	H	H	L	L	L	H	D ₇	D ₇
H	H	H	H	L	L	H	D _{7n}	D _{7n}

Notes: 1. H; High level, L; Low level, X; Irrelevant, Z; High impedance

Pin Arrangement



Recommended Operating Conditions

Item	Symbol	Ratings	Unit	Conditions
Supply voltage	V_{CC}	2 to 6	V	
Input / Output voltage	V_{IN}, V_{OUT}	0 to V_{CC}	V	
Operating temperature	T_a	-40 to 85	°C	
Input rise / fall time ^{*1}	t_r, t_f	0 to 1000	ns	$V_{CC} = 2.0\text{ V}$
		0 to 500		$V_{CC} = 4.5\text{ V}$
		0 to 400		$V_{CC} = 6.0\text{ V}$

Notes: 1. This item guarantees maximum limit when one input switches.

Waveform: Refer to test circuit of switching characteristics.

Electrical Characteristics

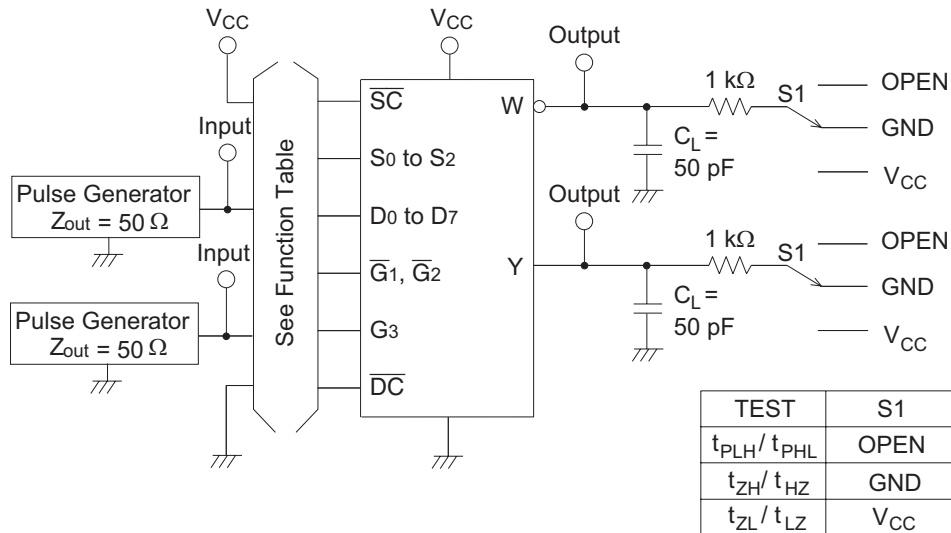
Item	Symbol	V_{CC} (V)	$T_a = 25^\circ\text{C}$			$T_a = -40\text{ to }+85^\circ\text{C}$		Unit	Test Conditions	
			Min	Typ	Max	Min	Max			
Input voltage	V_{IH}	2.0	1.5	—	—	1.5	—	V		
		4.5	3.15	—	—	3.15	—			
		6.0	4.2	—	—	4.2	—			
	V_{IL}	2.0	—	—	0.5	—	0.5	V		
		4.5	—	—	1.35	—	1.35			
		6.0	—	—	1.8	—	1.8			
Output voltage	V_{OH}	2.0	1.9	2.0	—	1.9	—	V	$V_{in} = V_{IH}\text{ or }V_{IL}$	$I_{OH} = -20\text{ }\mu\text{A}$
		4.5	4.4	4.5	—	4.4	—			$I_{OH} = -6\text{ mA}$
		6.0	5.9	6.0	—	5.9	—			$I_{OH} = -7.8\text{ mA}$
		4.5	4.18	—	—	4.13	—			
		6.0	5.68	—	—	5.63	—			
	V_{OL}	2.0	—	0.0	0.1	—	0.1	V	$V_{in} = V_{IH}\text{ or }V_{IL}$	$I_{OL} = 20\text{ }\mu\text{A}$
		4.5	—	0.0	0.1	—	0.1			
		6.0	—	0.0	0.1	—	0.1			
		4.5	—	—	0.26	—	0.33			$I_{OH} = 6\text{ mA}$
		6.0	—	—	0.26	—	0.33			$I_{OH} = 7.8\text{ mA}$
Off-state output current	I_{OZ}	6.0	—	—	± 0.5	—	± 5.0	μA	$V_{in} = V_{IH}\text{ or }V_{IL}$, $V_{out} = V_{CC}\text{ or GND}$	
Input current	I_{in}	6.0	—	—	± 0.1	—	± 1.0	μA	$V_{in} = V_{CC}\text{ or GND}$	
Quiescent supply current	I_{CC}	6.0	—	—	4.0	—	40	μA	$V_{in} = V_{CC}\text{ or GND}$, $I_{out} = 0\text{ }\mu\text{A}$	

Switching Characteristics

(C_L = 50 pF, Input t_r = t_f = 6 ns)

Item	Symbol	V _{CC} (V)	Ta = 25°C			Ta = -40 to +85°C		Unit	Test Conditions
			Min	Typ	Max	Min	Max		
Propagation delay time	t _{PLH} t _{PHL}	2.0	—	—	235	—	295	ns	D ₀ – D ₇ to output
		4.5	—	23	47	—	59		
		6.0	—	—	40	—	50		
	t _{PLH} t _{PHL}	2.0	—	—	270	—	340	ns	Data control to output
		4.5	—	26	54	—	68		
		6.0	—	—	46	—	58		
	t _{PLH} t _{PHL}	2.0	—	—	285	—	355	ns	S ₀ – S ₂ to output
		4.5	—	26	57	—	71		
		6.0	—	—	48	—	60		
	t _{PLH} t _{PHL}	2.0	—	—	300	—	375	ns	Select control to output
		4.5	—	26	60	—	75		
		6.0	—	—	51	—	64		
Output enable time	t _{ZH} t _{ZL}	2.0	—	—	150	—	190	ns	
		4.5	—	12	30	—	38		
		6.0	—	—	26	—	33		
Output disable time	t _{LZ} t _{HZ}	2.0	—	—	165	—	205	ns	
		4.5	—	15	33	—	41		
		6.0	—	—	28	—	35		
Setup time	t _{su}	2.0	50	—	—	65	—	ns	D ₀ – D ₇ to Data control, S ₀ – S ₇ to Select control
		4.5	10	2	—	13	—		
		6.0	10	—	—	13	—		
Hold time	t _h	2.0	5	—	—	5	—	ns	D ₀ – D ₇ to Data control, S ₀ – S ₇ to Select control
		4.5	5	0	—	5	—		
		6.0	5	—	—	5	—		
Pulse width	t _w	2.0	80	—	—	100	—	ns	Select control or Data control
		4.5	16	4	—	20	—		
		6.0	14	—	—	17	—		
Output rise/fall time	t _{TLH} t _{THL}	2.0	—	—	60	—	75	ns	
		4.5	—	4	12	—	15		
		6.0	—	—	10	—	13		
Input capacitance	C _{in}	—	—	5	10	—	10	pF	

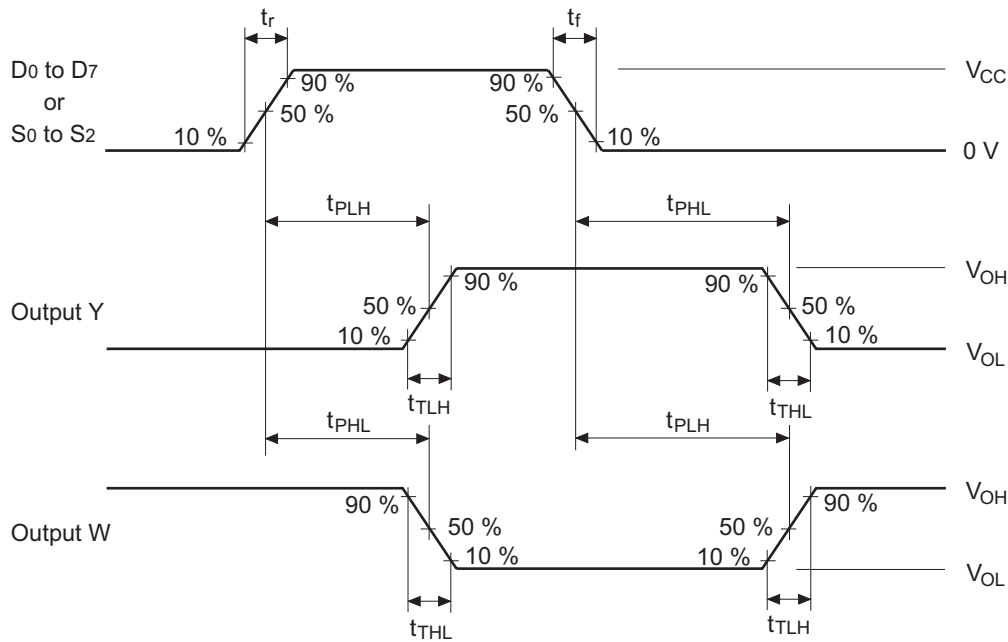
Test Circuit



Note : 1. C_L includes probe and jig capacitance.

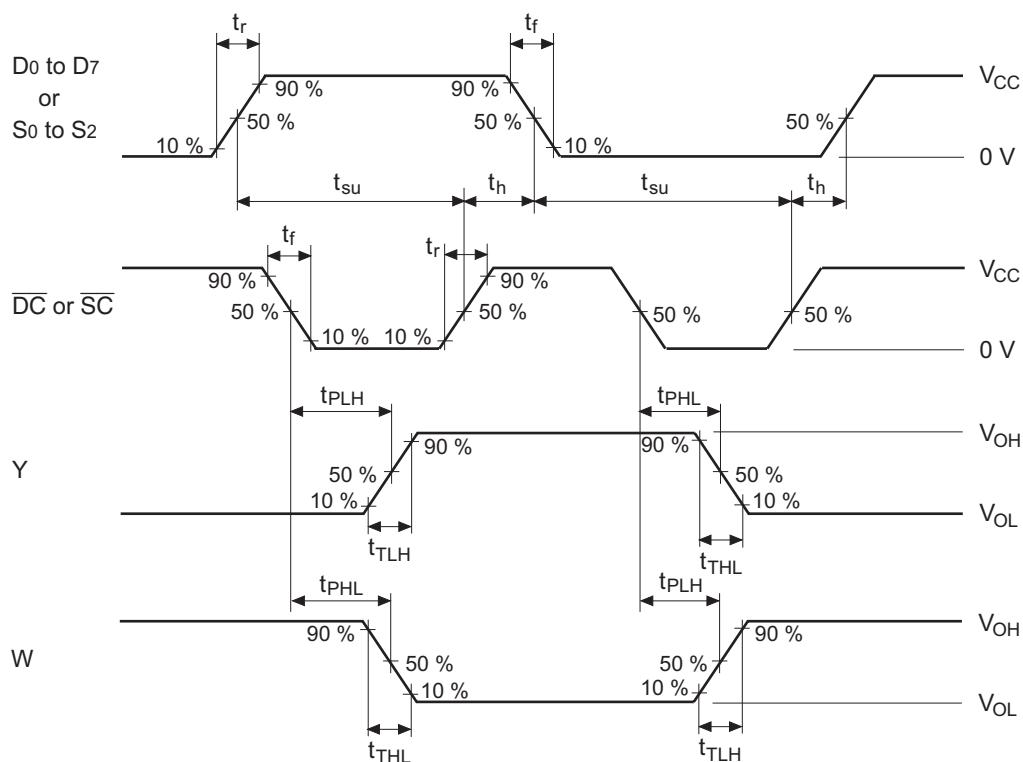
Waveforms

• Waveform – 1



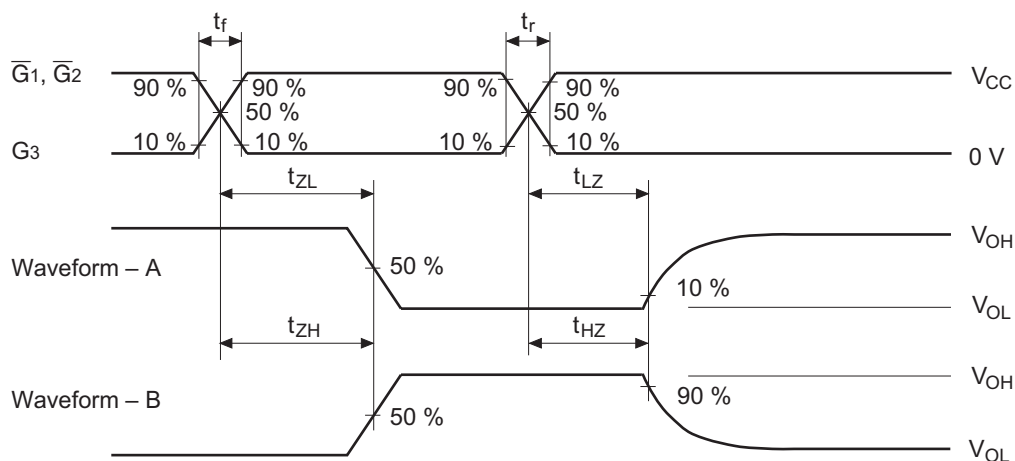
Notes: 1. Input waveform: $PRR \leq 1 \text{ MHz}$, $Z_o = 50 \Omega$, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$

• Waveform – 2



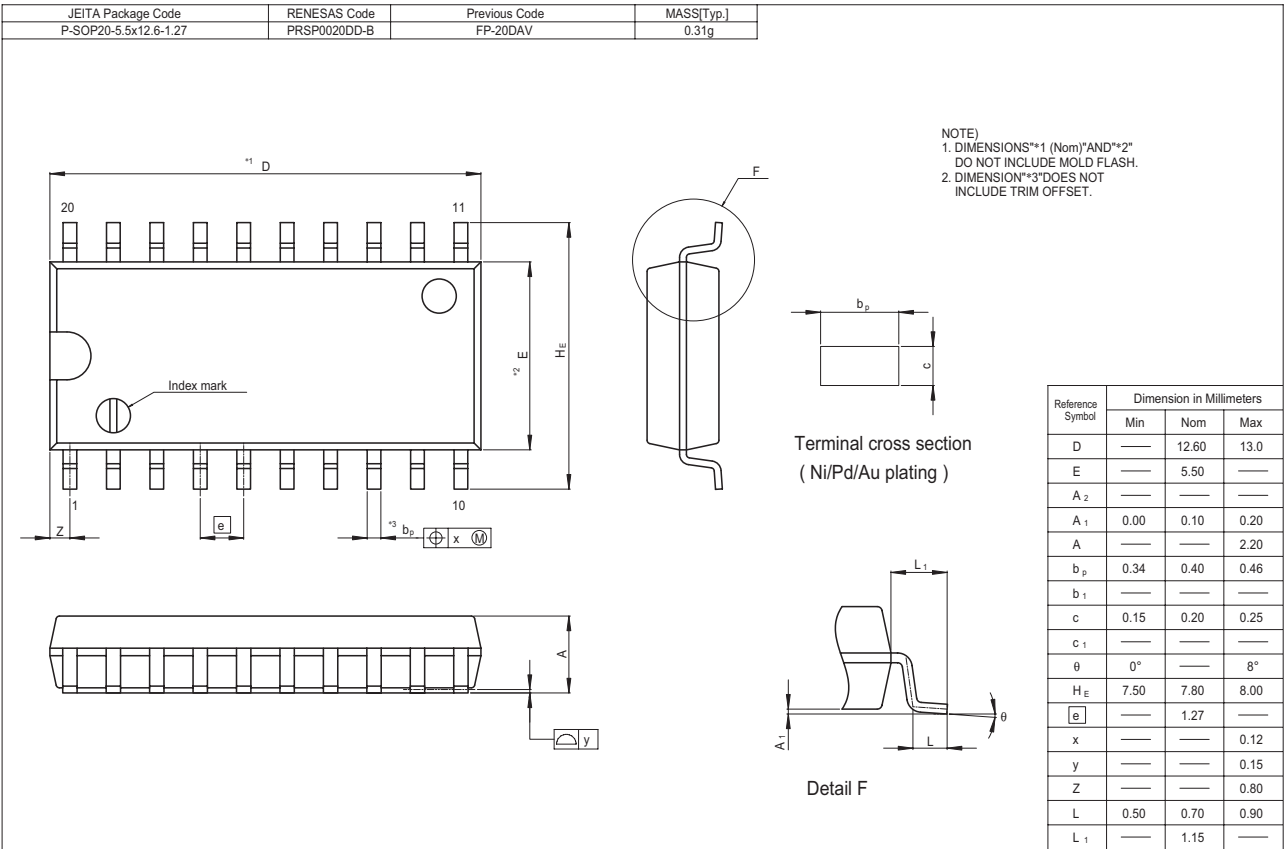
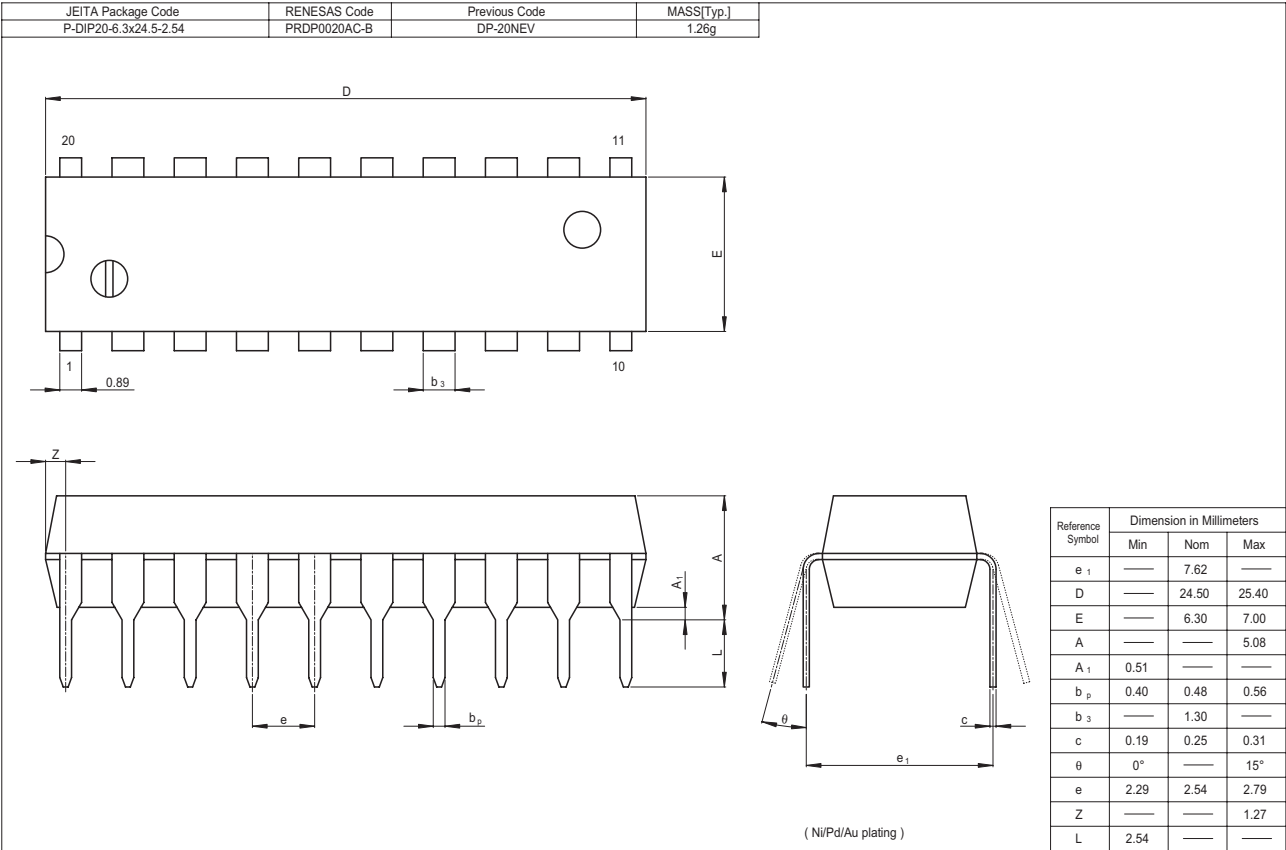
Notes: 1. Input waveform: $PRR \leq 1 \text{ MHz}$, $Z_o = 50 \Omega$, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$

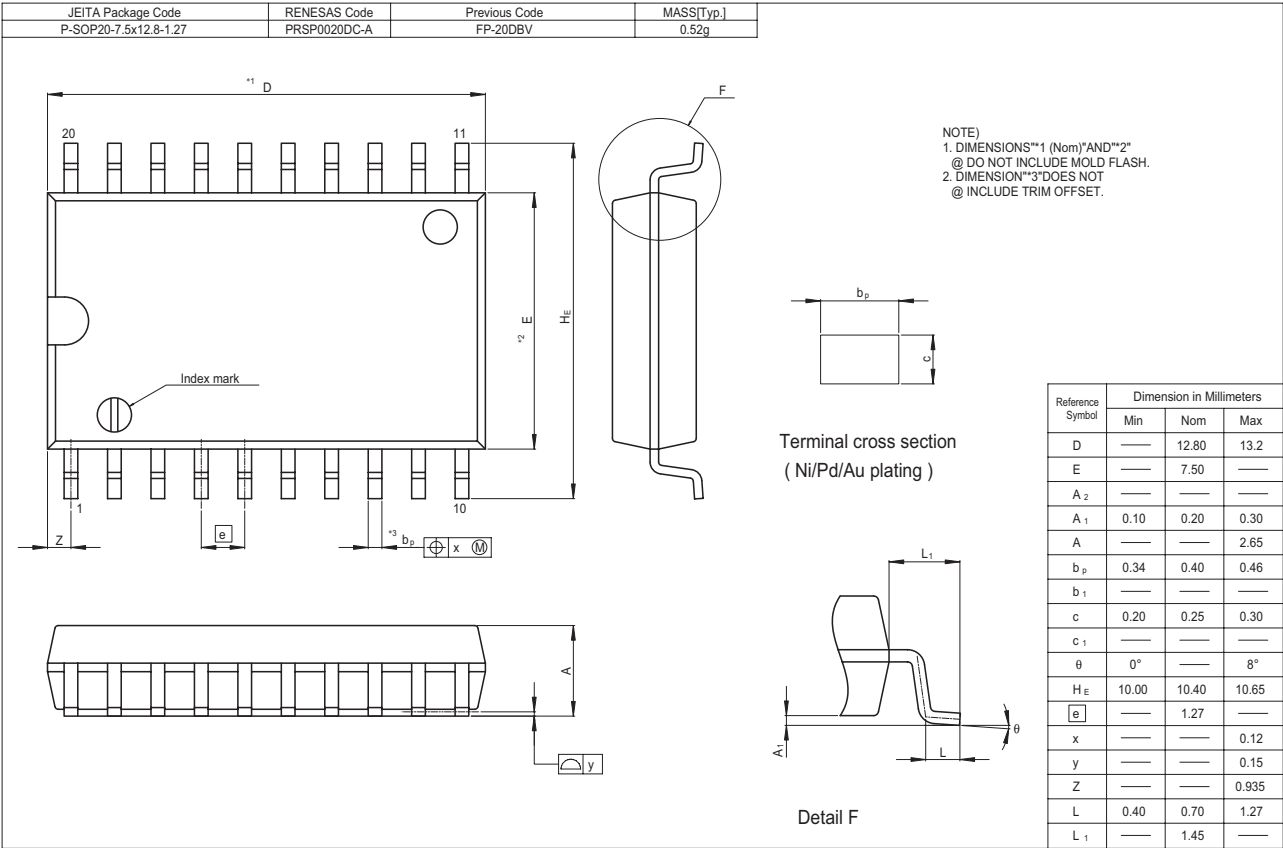
• Waveform – 3



Notes : 1. Input waveform : $PRR \leq 1 \text{ MHz}$, duty cycle 50%, $t_r \leq 6 \text{ ns}$, $t_f \leq 6 \text{ ns}$
 2. Waveform– A is for an output with internal conditions such that the output is low except when disabled by the output control.
 3. Waveform– B is for an output with internal conditions such that the output is high except when disabled by the output control.
 4. The output are measured one at a time with one transition per measurement.

Package Dimensions





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